A Quantity Theory of Money Approach to Cryptocurrency Valuation: Store of Value, Medium of Exchange, and the Economic Moat Signal

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Abstract

We apply the Quantity Theory of Money (QTM) to the cross-sectional valuation of digital assets, with an explicit distinction between two fundamentally different asset types. For native proof-of-stake coins—monetary instruments of decentralized blockchain networks—locking supply into validation contracts creates a Store of Value (SoV) premium distinct from the Medium of Exchange (MoE) component. The SoV/MoE ratio simplifies to $\lambda/(1-\lambda)$, the ratio of locked to freely circulating supply, observable on-chain without any velocity estimation. For governance and DeFi protocol tokens, no SoV/MoE decomposition applies because these assets lack a medium-of-exchange function. Instead, λ—the fraction of supply voluntarily committed to

governance or protocol staking—is an economic moat signal: holders who forgo liquidity to govern a protocol reveal fundamental conviction in its growth through a costly, credible action. In both cases λ positively predicts cross-sectional returns; the predictive power is concentrated among assets that are simultaneously cheap relative to their fundamental economic activity. We condition on the Growth-Levelized NVT ratio (market capitalization relative to discounted expected on-chain volume, NVT_GL) for coins, and on the NV/TVL ratio (market capitalization relative to Total Value Locked) for tokens. Across a panel of 160 assets from March 2020 through May 2026, a long-Stars (high λ, low valuation ratio) / short-Avoid (low λ, high valuation ratio) portfolio generates significant factor-adjusted alpha after controlling for market, size, and momentum.

1 Introduction

The cryptocurrency and DeFi ecosystem contains two economically distinct types of digital asset that require different valuation frameworks. *Native blockchain coins*—ETH, SOL, ADA, TRX, and their peers—are the monetary instruments of decentralized computing networks. Their value is governed, to a first approximation, by the same forces that determine the value of any currency: the scale of economic activity conducted on the network and the efficiency with which the coin circulates. The correct framework is the Quantity Theory of Money (Fisher, 1911). *Governance and DeFi protocol tokens*—UNI, AAVE, CRV, LDO, and hundreds of smart-contract-issued assets—are the governance and incentive instruments of decentralized financial protocols. Their value reflects the productive capacity of the protocol, not coin velocity. The correct framework for these assets is a governance moat model. This paper develops both frameworks, derives a common conviction signal λ applicable across both asset types, and shows that λ positively predicts cross-sectional returns in each. The mechanics of how λ generates that predictability, and how it should be measured and conditioned upon, differ across the two tracks.

For native coins, the starting point is the Fisher equation, $MV = PQ$: market capitalization (M, the money supply proxy) times velocity (V) equals nominal on-chain output (PQ), the annualized value of economic transactions conducted on the network. Not all circulating supply is economically active, however. A growing fraction is locked in proof-of-stake validation contracts, pledged as governance collateral, or held dormant with near-zero velocity. These holdings perform no current transactional function: they contribute to money supply but not to exchange. Separating economically active from inert supply, we show that market capitalization decomposes into a Medium of Exchange (MoE) component, priced by the velocity of unlocked supply, and a Store of Value (SoV) premium, reflecting holder conviction about future network growth (Sockin and Xiong, 2020; Krishnamurthy and Vissing-Jorgensen, 2012). The formal derivation in Appendix A establishes that this

SoV/MoE ratio simplifies to

$$\frac{\mathrm{MC}_{\mathrm{SoV}}}{\mathrm{MC}_{\mathrm{MoE}}} = \frac{\lambda}{1-\lambda}, \tag{1}$$

where λ is the fraction of total supply locked or staked. This ratio is directly observable from on-chain data without any velocity estimation. For PoS coins, locking is grounded in rational optimization: seigniorage distributed to stakers constitutes newly issued monetary income (Cong and He, 2024), and the staking decision reveals that this income plus anticipated price appreciation exceeds the opportunity cost of forgoing liquidity. The SoV premium is thus the present value of expected future convenience yields anticipated by committed holders (Krishnamurthy and Vissing-Jorgensen, 2012; Nagel, 2016). Monetary search theory (Kiyotaki and Wright, 1993; Lagos and Wright, 2005) further implies that high λ is self-reinforcing: the same coordination dynamics that make money valuable make a high-λ holder base a credible signal of network adoption. For PoW coins (BTC, LTC, and others without a validator staking mechanism), the SoV/MoE decomposition applies in principle, but λ can only be approximated through supply dormancy—the HODL-wave approach—rather than staking. We discuss this separately in Section 3.

For governance tokens, the SoV/MoE decomposition does not apply. A governance token such as UNI does not need to change hands to facilitate Uniswap trades; it confers governance rights, not transactional capacity. There is no MoE component to separate from an SoV premium: the QTM velocity framework is not the right model. What λ signals for tokens is instead *governance conviction*. When a holder locks tokens into a DAO contract or delegates voting power, they incur a real cost—illiquidity, monitoring effort, foregone trading flexibility—and by doing so they reveal, through a costly and observable action, a fundamental belief in the protocol's future that passive holders do not share. Drawing on Spence (1973), the costliness of the action is what makes it credible. Drawing on Hirschman (1970) and Edmans and Manso (2011), exercising voice (governance participation) rather than exit (selling) is the behavioral marker of committed, informed principals. The fraction of supply committed to governance is therefore an economic moat indicator: protocols with high λ have cultivated a holder base sufficiently convinced of their growth trajectory to pay the participation cost.

Despite the different economic mechanisms, the empirical structure of the conviction signal is common to both asset types. In both cases, aggregate locking behavior reveals information about the asset's future trajectory that price does not yet fully reflect: for coins because extracting the signal from λ requires knowing the distribution of private costs and signal noise—not common knowledge—and for tokens because governance participation data are read by few market participants. Both tracks motivate a partial-revelation story in the spirit of Grossman and Stiglitz (1980): the conviction signal is not costlessly impounded into price, so high-λ assets systematically outperform as the signal is gradually validated. The conditioning variable measures how much of the signal the market has already absorbed. For coins, this is the Growth-Levelized NVT ratio, NVT_GL = MC/PQ^*, where PQ^* is the discounted present value of expected on-chain transaction volume, growth-adjusted analogously to a DCF model. For tokens, this is the NV/TVL ratio, market capitalization divided by Total Value Locked. Low NVT_GL for a coin, or low NV/TVL for a token, indicates that the market has not yet priced in the conviction signal embedded in λ. Assets combining high λ with low valuation ratio—*Stars* in our quadrant—offer the best expected returns; those combining low λ with high valuation ratio—*Avoid*—offer the worst. Our main empirical result is that this quadrant framework delivers positive and significant factor-adjusted alpha across both the coin and token subsamples.

Related Literature. A growing body of work attempts to value digital assets, and our paper connects to several strands. On equilibrium cryptocurrency valuation, Pagnotta and Buraschi (2022) model Bitcoin value as a function of user adoption and network security; Biais, Bisière, Bouvard, and Casamatta (2019) study equilibrium selection in blockchain protocols; Schilling and Uhlig (2019) develop a general equilibrium model of Bitcoin as a competing currency; and Easley, O'Hara, and Basu (2019) analyze how transaction fees connect mining incentives to market value. Cong, Li, and Wang (2021) and Sockin and Xiong (2020) are the closest to our approach in modeling token prices through adoption dynamics and convenience yields respectively; we complement them by providing an empirically tractable QTM decomposition and a testable investment signal derived from on-chain observables. On crypto risk factors and return predictability, Liu and Tsyvinski (2021) identify network, momentum, and investor attention factors; Liu, Tsyvinski, and Wu (2022) develop a three-factor model for cryptocurrency returns; and Makarov and Schoar (2020) document price dislocations across exchanges. We contribute to this literature the first on-chain supply-side signal—$\lambda/(1-\lambda)$—grounded in monetary theory rather than return-based factor construction. On the monetary economics of digital assets, Kiyotaki and Wright (1993), Lagos and Wright (2005), and Schilling and Uhlig (2019) provide search-theoretic and general equilibrium foundations; Cong and He (2024) analyzes proof-of-stake tokenomics; and Jiang, Krishnamurthy, and Lustig (2024) draw the analogy between dollar exorbitant privilege and dominant-network premia. The convenience yield literature (Krishnamurthy and Vissing-Jorgensen, 2012; Nagel, 2016) motivates the monetary premium in $\mathrm{MC}_{\mathrm{SoV}}$. On token issuance and governance, Howell, Niessner, and Yermack (2020) study the determinants of ICO success; Chod and Lyandres (2021) model ICO design under information asymmetry; and Edmans and Manso (2011) and Hirschman (1970) provide the governance theory linking holder participation to firm and protocol value. Our paper is the first, to our knowledge, to derive a directly observable valuation decomposition from QTM, to distinguish the signal content of λ for coins versus governance tokens, and to demonstrate that the interaction of conviction and valuation predicts cross-sectional cryptocurrency returns.

The remainder of the paper is organized as follows. Section 2 develops the theoretical frameworks for coins (SoV/MoE) and tokens (governance moat) and derives the testable propositions. Section 3 defines and constructs the three key variables: the conviction index λ, the Growth-Levelized NVT ratio for coins, and the NV/TVL ratio for tokens. Section 4 describes the data sources, sample construction, and summary statistics. Section 5 presents the cross-sectional return predictability and portfolio results. Section 6 reports robustness tests. Section 7 concludes.

2 Theoretical Framework

This section develops the theoretical argument in three steps. Section 2.1 establishes the coin framework: QTM applied to native blockchain coins yields a SoV/MoE decomposition in which the locking ratio λ is the sufficient statistic. Section 2.2 develops the token framework: governance tokens lack a medium-of-exchange function, so λ is reinterpreted as a governance moat signal grounded in costly signaling theory. Section 2.3 introduces the common valuation-conditioning mechanism that applies across both tracks. Each subsection closes with the empirical propositions we test.

2.1 The Coin Framework: SoV/MoE and the Staking Ratio

The QTM identity applied to a native blockchain coin treats market capitalization as money supply: $\mathrm{MC} \cdot V = PQ$, where V is the velocity of supply in transactional use and PQ is nominal on-chain output. The critical observation is that supply is heterogeneous in velocity. A fraction λ of total circulating supply is locked in proof-of-stake validation contracts, staked to earn seigniorage, or otherwise immobilized with near-zero velocity. The complementary fraction $(1-\lambda)$ is freely circulating and economically active. Only the active fraction supports transactions; the locked fraction is held as a store of value and does not circulate. Partitioning supply accordingly, the appendix shows that market capitalization decomposes as

$$\mathrm{MC} = \underbrace{\frac{PQ}{V_{\text{active}}}}_{\mathrm{MC}_{\mathrm{MoE}}} + \underbrace{\frac{PQ}{V_{\text{active}}} \cdot \frac{\lambda}{1-\lambda}}_{\mathrm{MC}_{\mathrm{SoV}}},$$

and therefore the SoV/MoE ratio satisfies

$$\frac{\mathrm{MC}_{\mathrm{SoV}}}{\mathrm{MC}_{\mathrm{MoE}}} = \frac{\lambda}{1-\lambda}.$$

This is an accounting identity given λ, not yet a behavioral prediction.

The behavioral prediction follows from modeling the individual staking decision. Consider a continuum of agents indexed by $i \in [0,1]$, each choosing at date t whether to stake one unit of the native coin or hold it freely circulating. Staking yields a publicly observed current reward $b_t > 0$—seigniorage paid to validators in a PoS network (Cong and He, 2024)—plus a private convenience yield $c_{i,t+1} = \theta_{t+1} + \varepsilon_i$, where θ_{t+1} is the network's true future growth trajectory (unobserved at t) and ε_i is mean-zero idiosyncratic signal noise. Staking also imposes a private liquidity cost φ_i (i.i.d., independent of θ_{t+1} and ε_i). Agent i stakes if and only if

$$b_t + \theta_{t+1} + \varepsilon_i \geq \varphi_i \quad \Longleftrightarrow \quad \eta_i \leq b_t + \theta_{t+1}, \qquad \eta_i \equiv \varphi_i - \varepsilon_i.$$

By the law of large numbers, the aggregate staking ratio converges almost surely to

$$\lambda_t = H(b_t + \theta_{t+1}),$$

where H is the CDF of η_i. Idiosyncratic cost and signal noise wash out in the cross-section; what survives is a deterministic, strictly increasing function of θ_{t+1}. Because b_t is publicly observed, λ_t is a publicly verifiable signal of the holder base's aggregate private conviction about the network's future growth. Extracting θ_{t+1} from λ_t, however, requires knowing H and b_t with precision—not common knowledge for most market participants. As in Grossman and Stiglitz (1980), the signal is not fully impounded into price at t. As θ_{t+1} materializes and becomes public over $t+1$, price converges toward the value anticipated by high-λ stakers, generating positive return predictability.

Proposition 1a (Coins). $\mathrm{Cov}_j(r_{j,t+1},\, \lambda_{j,t}/(1-\lambda_{j,t})) > 0$ in the cross-section: coins with higher SoV/MoE ratios earn higher subsequent returns, after controlling for size, momentum, and market beta.

Hypothesis 1a. *For native proof-of-stake coins, assets with higher* $\lambda/(1-\lambda)$ *in month* t *earn statistically and economically significant higher returns over month* $t+1$*, after controlling for coin-level size, momentum, and market exposure. The result holds regardless of whether the coin is an L1 or L2, provided the staking mechanism confers a positive seigniorage benefit* $b_t > 0$*.*

2.2 The Token Framework: Governance Conviction as Economic Moat

For governance and DeFi protocol tokens, the SoV/MoE framework is inapplicable: these tokens are not media of exchange. A holder of UNI does not need to spend UNI to use Uniswap; holding governance tokens does not reduce the protocol's transactional capacity. The token has no meaningful velocity in the QTM sense, and the SoV/MoE decomposition carries no economic content. What λ signals for tokens is something different: *governance conviction*—the fraction of supply whose holders have voluntarily incurred a cost to demonstrate fundamental belief in the protocol's future.

The formal model is a straightforward extension of the coin model. The same continuum of agents now hold a governance token at date t and choose whether to lock it into a governance contract (vote-escrow, delegation, or protocol staking) or hold it freely. Locking carries a private cost $\varphi_i = \ell + \kappa_i$, decomposed into a common illiquidity cost $\ell \geq 0$ (positive for vote-escrow protocols such as Curve's veCRV or Balancer's veBAL, zero for simple delegation under ERC20Votes) and an idiosyncratic engagement cost $\kappa_i \geq 0$, the monitoring and time cost of exercising governance voice rather than exit (Hirschman, 1970; Edmans and Manso, 2011). The current locking benefit $b_t \geq 0$ may include a fee or revenue share for protocols with on-chain distribution; for most governance tokens $b_t = 0$ and the incentive to lock is purely forward-looking. The threshold condition is identical to the coin case: $\eta_i = \varphi_i - \varepsilon_i \leq b_t + \theta_{t+1}$, yielding $\lambda_t = H(b_t + \theta_{t+1})$.

The key difference is interpretation. For coins, $b_t > 0$ is paid in newly minted seigniorage, so staking is partly a rational income decision. For tokens with $b_t = 0$, locking is a purely conviction-revealing action: the holder incurs $\ell + \kappa_i$ with no current compensation, revealing that their private belief about θ_{t+1} is strong enough to make the commitment worthwhile. This is a Spence (1973) costly-signaling equilibrium: only holders with sufficiently high conviction rationally pay the cost. The resulting aggregate λ_t is therefore a cleaner signal of fundamental conviction for governance tokens than for coins, because the seigniorage confound is absent. The partial-revelation logic applies identically: the signal in λ_t is not freely extractable from public data, so it is not instantly priced, generating the same return predictability.

Proposition 1b (Tokens). $\mathrm{Cov}_j(r_{j,t+1}, \lambda_{j,t}) > 0$ in the cross-section of governance tokens: assets with higher governance conviction ratios earn higher subsequent returns, after controlling for size, momentum, and protocol category.

Hypothesis 1b. *For DeFi governance tokens, assets with higher λ in month t earn statistically and economically significant higher returns over month $t+1$, after controlling for token-level size, momentum, and protocol category. The predictive power of voting-participation-weighted λ exceeds that of passive locking alone, consistent with the costly-signaling logic: the harder the action, the purer the signal.*

2.3 The Valuation Conditioning Signal

Propositions 1a and 1b establish that λ carries information not yet fully in price. What they do not characterize is how much of the signal remains unpriced at a given date. Let $\delta_t \in [0, 1]$ denote the fraction of the conviction signal in λ_t already absorbed into the current price, so that $p_t = p_t^{\text{fund}}(1 + \delta_t)$ where p_t^{fund} reflects current economic activity. The expected return as the signal is validated over $t + 1$ is proportional to the unabsorbed portion $\lambda_t(1 - \delta_t)$, not to λ_t alone. The marginal predictive power of λ_t is therefore decreasing in δ_t, and δ_t is increasing in the current market capitalization relative to fundamental activity.

For coins, the relevant fundamental is discounted expected on-chain volume: δ_t is increasing in $\text{NVT\_GL}_t = \text{MC}_t / PQ^*_t$, where PQ^*_t is the growth-adjusted present value of expected transaction activity (formally defined in Section 3.2). For tokens, the relevant fundamental is Total Value Locked: δ_t is increasing in $\text{NV/TVL}_t = \text{MC}_t/\text{TVL}_t$. In both cases, a high ratio indicates that current market capitalization already embeds the conviction signal, and the incremental return predictability of λ_t is low; a low ratio indicates the signal remains unabsorbed.

Proposition 2. *The cross-partial effect of λ_t and the valuation ratio on expected returns is negative: the return predictability of λ_t is concentrated among low-valuation-ratio assets.*

Hypothesis 2. *The return predictability documented in H1a and H1b is significantly stronger among assets with below-median valuation ratios (NVT_GL for coins, NV/TVL for tokens). The marginal return to a unit increase in λ is higher when the asset is cheap relative to its fundamental economic activity.*

The valuation ratio carries no independent return prediction in this framework. A low NVT_GL alone—without a high λ—merely describes a coin with limited current throughput relative to market cap, not necessarily a mispricing; similarly for NV/TVL. The conditioning role is asymmetric: valuation identifies the *timing* of conviction signal absorption, not a separate source of alpha.

2.4 Hypotheses and the Quadrant Portfolio

Propositions 1 and 2 jointly define a two-dimensional framework. *Stars* combine high λ with low valuation ratio—strong conviction, signal not yet priced—and are predicted to have the highest future returns. *Avoid* assets combine low λ with high valuation ratio—weak conviction, valuation already stretched—and are predicted to have the lowest returns.

Hypothesis 3. *A long-short portfolio that goes long Star assets (above-median λ, below-median valuation ratio) and short Avoid assets (below-median λ, above-median valuation* *ratio), rebalanced monthly, generates positive and statistically significant factor-adjusted alpha after controlling for market, size, and momentum. The result holds separately within the coin sample (using NVT_GL) and within the token sample (using NV/TVL), and in the combined panel. The Stars-minus-Avoid Sharpe ratio exceeds that of equal-weighted benchmark portfolios.*

H3 is the integrating test: it asks whether the joint two-dimensional signal has incremental predictive power beyond either dimension alone. We report results separately for coins and tokens to preserve the track distinction: for coins, alpha reflects the return to SoV/MoE mispricing correcting over time; for tokens, alpha reflects the market's underreaction to governance conviction signals.

3 Variable Construction

The theoretical framework in Section 2 requires three empirical proxies: the conviction index λ, common to both asset tracks; the Growth-Levelized NVT ratio for the coin track; and the NV/TVL ratio for the token track. This section defines each variable and describes its construction. Data sources and sample formation are covered in Section 4.

3.1 The Conviction Index λ

The theoretical λ_t is the aggregate locking or commitment ratio of an asset's holder base at date t. Empirically, the conviction-revealing actions observable on-chain vary by asset type and protocol design. We therefore construct λ as a composite index over three conceptually motivated channels, and aggregate across available channels using cross-sectional z-scores.

Channel 1: Staking. For proof-of-stake layer-one coins, the staking channel is the fraction of circulating supply bonded as validator or delegator stake at month-end. This is the direct empirical counterpart of the locking action in Section 2.1: it simultaneously earns seigniorage ($b_t > 0$) and constitutes a costly, publicly visible commitment to the network's future. Staking rates are sourced from chain-native archive data for each coin.[1] The staking channel covers 48 assets and 2,222 asset-months.

Channel 2: Holding. The holding channel measures the fraction of circulating supply that has not transacted on-chain in the preceding six months (HODL-6m). This proxy extends conviction measurement to any EVM-compatible asset without a formal staking mechanism, mapping to the inert supply fraction λ_t in the model: a holder who has not transacted for six months has revealed a preference for holding over liquidity. HODL-6m is computed by replaying on-chain Transfer event logs in chronological order under a FIFO attribution rule, assigning each unit of supply to its most recent transfer date.[2] This channel spans 408 assets and 11,654 asset-months, providing the broadest coverage in the conviction panel.

Channel 3: Governance. Two governance sub-channels capture active participation in protocol decision-making. The delegation sub-channel records the fraction of circulating supply delegated to a voting representative under an ERC20Votes contract, observable from on-chain Delegate event logs.[3] The voting sub-channel records proposal participation rates from Snapshot off-chain governance records, cross-referenced with on-chain ERC20Votes logs where available. Both sub-channels identify holders who have incurred the engagement costs

[1] For Polkadot (DOT) and Kusama (KSM), bonded supply is retrieved from the Subscan archive API. For Cosmos-ecosystem chains (CRO, KAVA, OSMO), we query the Cosmos SDK Staking module endpoint at block heights corresponding to month-end timestamps, with timestamps identified via binary search on the CometBFT RPC interface. For EVM-based staking contracts (BNB, MATIC, TRX, and others), staked supply is derived from on-chain staking event logs via the Etherscan multi-chain archive API. Each data source is validated by confirming that reported values vary across block heights, ensuring the archive reflects historical rather than current-only state.

[2] Transfer events are retrieved from the Etherscan Pro V2 multi-chain archive, covering Ethereum mainnet and twelve additional EVM chains. Two data quality filters are applied: (i) monthly on-chain flow exceeding 100× reported circulating supply is classified as a minting, bridging, or aggregator artifact and excluded; (ii) asset-months in which HODL-6m exceeds 80% persistently are flagged for robustness checks, consistent with possible custodial cold-wallet concentration rather than individual holder conviction. Self-transfers (sender equals recipient) are excluded from the event replay.

[3] Data sourced from `DelegateVotesChanged` event logs via the Etherscan Pro V2 API, aggregated to month-end totals.

κ_i of Section 2.2—the costlier the action, the more selected and informative the resulting signal. The delegation and voting sub-channels cover 26 and 53 assets, respectively, concentrated in the DeFi governance token sample.

Composite Index. For each asset-month, each available channel is standardized to a z-score over all asset-months with non-missing observations for that channel. The composite λ_z is the equal-weight average of standardized channel scores available for that asset-month. The result has approximately zero cross-sectional mean and is defined whenever at least one channel is observed. Equal weighting imposes no prior on the relative information content of the channels; channel-specific predictive power is examined in robustness tests. The full conviction panel covers 467 assets and 13,626 asset-months, of which 11,901 are single-channel, 1,371 two-channel, and 354 three-or-more-channel observations.

3.2 Growth-Levelized NVT

For the coin regression track, the valuation anchor is the Growth-Levelized Network Value to Transactions ratio (NVT_GL), which scales market capitalization by the growth-adjusted, risk-discounted present value of expected on-chain throughput:

$$\text{NVT\_GL}_{j,t} = \frac{\text{MC}_{j,t}}{PQ^*_{j,t}}, \tag{2}$$

$$PQ^*_{j,t} = \frac{\displaystyle\sum_{s=1}^{n} \frac{PQ_{0,j,t}(1+g_{j,t})^s}{(1+r_{e,j,t})^s} + \frac{PQ_{0,j,t}(1+g_{j,t})^n(1+g_\infty)}{(r_{e,j,t}-g_\infty)(1+r_{e,j,t})^n}}{\dfrac{1-(1+r_{e,j,t})^{-n}}{r_{e,j,t}}}, \tag{3}$$

where $PQ_{0,j,t}$ is the trailing twelve-month annualized on-chain transfer volume for asset j at month t; $g_{j,t}$ is the trailing three-year compound annual growth rate of PQ_0, capped in $[-50\%, 200\%]$; $n = 10$ years is the explicit-growth projection horizon; $g_\infty = 3\%$ is the terminal perpetuity growth rate; and $r_{e,j,t} = r_f + \hat{\beta}_{j,t} \cdot \text{MRP}$ is the asset-specific discount rate, with $r_f = 4\%$, $\text{MRP} = 30\%$, and $\hat{\beta}_{j,t}$ the trailing 36-month beta against the BTC price index, floored so that $r_{e,j,t} \geq g_\infty + 2\%$ to ensure well-defined terminal values.[4]

NVT_GL below one signals that market capitalization is below the growth-adjusted present value of expected on-chain activity. NVT_GL substantially above one indicates that current market capitalization already embeds forward growth not yet visible in throughput. The growth adjustment corrects a systematic bias in raw NVT: fast-growing assets may appear expensive on a raw NVT basis but inexpensive once expected growth is discounted.

3.3 NV/TVL Ratio

For the token regression track, the valuation anchor is the Network Value to Total Value Locked ratio:

$$\text{NV/TVL}_{j,t} = \frac{\text{MC}_{j,t}}{\text{TVL}_{j,t}},$$

where $\text{TVL}_{j,t}$ is the market value of assets deposited in protocol j's smart contracts at month-end t. TVL proxies for the scale of economic activity a protocol intermediates—the token analog of on-chain transaction volume in the QTM framework—and is the natural denominator for a price-to-fundamentals ratio in DeFi. Where a protocol is deployed across multiple chains, we use the sum of all-chain TVL.

A low NV/TVL indicates that market capitalization is modest relative to the economic activity under the protocol's control. A high NV/TVL indicates that market capitalization substantially exceeds the protocol's current economic footprint. Unlike NVT_GL, NV/TVL is not growth-adjusted: TVL is itself a current-period flow variable that captures realized protocol adoption, and a separate growth correction would risk double-counting the forward information already embedded in TVL levels.

[4] The denominator of (3) is the standard annuity factor normalizing the finite-horizon present value of future PQ to an equivalent annualized flow, making NVT_GL interpretable as a price-to-normalized-activity ratio analogous to a price-to-normalized-earnings multiple in equity analysis. The MRP of 30% reflects the elevated required return on cryptocurrency risk; $\hat{\beta}_{j,t}$ is retained in the panel so alternative discount rate assumptions can be applied directly in robustness tests.

4 Data

4.1 Data Sources

Price, market capitalization, and circulating supply are drawn from CoinMarketCap at daily granularity, snapped to end-of-calendar-month observations. CoinMarketCap provides the broadest historical coverage of digital asset market data and is the standard reference for universe construction in the cryptocurrency finance literature (Liu, Tsyvinski, and Wu, 2022; Makarov and Schoar, 2020).

On-chain Transfer event logs, staking event logs, and ERC20Votes Delegate event logs are retrieved from the Etherscan Pro V2 multi-chain API, which provides archive access to EVM-compatible chains including Ethereum mainnet, BNB Chain, Polygon, Avalanche, Arbitrum, Optimism, and nine additional networks. Non-EVM staking data follows chain-native sources: Subscan archive API for Polkadot and Kusama; Cosmos SDK Staking module endpoints for Cosmos-ecosystem chains. Off-chain governance participation is sourced from the Snapshot API, cross-referenced with on-chain ERC20Votes logs.

On-chain transaction volume for the NVT_GL construction is assembled from three sources in priority order. DeFiLlama chain-level DEX volume[5] serves as the primary source for chains with active decentralized exchange infrastructure (1,340 asset-months). BitInfoCharts sent-in-USD settlement value[6] is used for PoW coins and pre-DEX periods, capturing all on-chain transfers rather than DEX activity alone (705 asset-months). Blockchain.com estimated transaction value[7] supplements Bitcoin and legacy chain observations with pre-DEX-era coverage (118 asset-months). The remaining 363 asset-months use DeFiLlama protocol-level volume for assets whose primary activity flows through specific protocols. Each series is validated against the chain's own reported aggregate, with the reconstructed monthly level required to lie within 5% of the chain total. The resulting NVT_GL panel

[5] `https://defillama.com`, accessed through May 2026.
[6] `https://bitinfocharts.com`, accessed through May 2026.
[7] `https://blockchain.com`, accessed through May 2026.

covers 67 assets and 2,526 valid asset-months from August 2016 through May 2026.

Total Value Locked is sourced from DeFiLlama at daily granularity, snapped to month-end; multi-chain protocols use the sum of all-chain TVL. The TVL panel covers 162 assets and 8,066 asset-months from September 2017 through May 2026.

4.2 Sample Construction

The starting universe is 1,939 digital assets drawn from the CoinMarketCap historical database, observed at end-of-calendar-month from August 2015 through May 2026, yielding 156,838 asset-month observations.[8]

Assets are classified into three mutually exclusive types: *coins* (native settlement assets of a public blockchain, e.g., BTC, ETH, SOL), *tokens* (smart-contract-issued assets deployed on a host chain, e.g., ERC-20 governance and DeFi tokens), and *other* (stablecoins, wrapped tokens, and infrastructure assets whose value is structurally indexed to an external reference or operating entity). Other-class assets are excluded from both regression samples because neither the SoV/MoE framework nor the governance moat framework applies to their valuation.

The two regression samples are defined by simultaneous availability of λ_z and a positive valuation denominator. The *coin regression sample* requires: (i) coin classification; (ii) proof-of-stake architecture; (iii) at least one month of non-missing λ_z; and (iv) at least one month of positive NVT_GL. This yields 24 assets and 718 asset-months spanning December 2020 through May 2026. The median NVT_GL in this sample is 0.017 and median market capitalization is $820 million. The *token regression sample* requires: (i) token classification; (ii) non-missing λ_z; and (iii) positive TVL. This yields 136 assets and 4,627 asset-months spanning March 2020 through May 2026. Median TVL is $55.8 million (IQR: $6.2M–$511.6M) and median market capitalization is $120 million.

[8]All assets with at least one non-zero price observation in CoinMarketCap are included. Market capitalization is computed as reported price times circulating supply. Continuous coverage is not required; assets may enter and exit the panel as they gain or lose listing.

Several populations are excluded from one or both samples by data structure. Proof-of-work coins have no staking mechanism and are therefore outside the scope of Hypothesis 1a, though they contribute to NVT_GL baseline tests. Among PoS coins, historical staking data is unavailable for chains whose archive nodes do not retain historical state, either due to aggressive pruning or reliance on third-party indexers for pre-recent-block queries. Among tokens, approximately 1,100 assets have no DeFiLlama TVL entry because they serve as utility, memetic, or infrastructure tokens for which protocol TVL is not defined. The remaining coverage gap arises from temporal non-overlap: tokens for which conviction data is available only after the period of active TVL reporting contribute to individual panels but not to the regression sample intersection. Channel 2 (holding) applies only to EVM-compatible assets; PoW coins with UTXO-based ledger accounting require a different pipeline based on UTXO age analysis and are left to future work.

The combined regression panel spans 160 unique assets and 5,345 asset-months. At the end of the sample in May 2026, the panel accounts for $465 billion in market capitalization—19.5% of the $2.38 trillion CoinMarketCap universe—consistent with the concentration of on-chain governance and staking infrastructure among the larger assets in the market.

4.3 Summary Statistics

Table 1 reports coverage across data panels and key statistics for the regression samples.

5 Results

[To be written.]

6 Robustness

[To be written.]

Table 1: Summary Statistics

Panel / Sample	Assets	Asset-months	Date range	Median MC
Full universe	1,939	156,838	2015-08 to 2026-05	\$6.5M
Conviction panel (λ_z)	467	13,626	2016-12 to 2026-05	—
ch1 staking	48	2,222	2017-07 to 2026-05	—
ch2 holding	408	11,654	2015-08 to 2026-05	—
ch3 delegation	26	627	2020-05 to 2026-05	—
ch3 voting	53	1,202	2020-08 to 2026-05	—
NVT_GL panel	67	2,526	2016-08 to 2026-05	—
TVL panel	162	8,066	2017-09 to 2026-05	—
Coin regression (NV/PQ*)	24	718	2020-12 to 2026-05	\$820M
Token regression (NV/TVL)	136	4,627	2020-03 to 2026-05	\$120M
Combined	160	5,345	2020-03 to 2026-05	\$147M

Notes: Median MC is the cross-sectional median market capitalization across all asset-months with a positive price observation. Conviction panel sub-channel counts sum to more than the panel total because 1,725 asset-months appear in two or more channels. Coin regression requires simultaneous non-missing λ_z and positive NVT_GL for proof-of-stake coins. Token regression requires simultaneous non-missing λ_z and positive TVL. Median NVT_GL in the coin regression sample is 0.017. Median TVL in the token regression sample is \$55.8M (IQR: \$6.2M–\$511.6M).

7 Conclusion

[To be written.]

A Derivation of the SoV/MoE Decomposition

This appendix provides the formal derivation of the SoV/MoE decomposition and its simplification to a function of λ alone.

Setup. Let M denote the total circulating supply of a digital asset and MC its market capitalization, which serves as the money supply proxy in the QTM framework. Let PQ denote nominal economic output (on-chain transaction volume for coins; protocol throughput for tokens). Standard QTM gives $\text{MC} \cdot V = PQ$, so the market capitalization satisfies $\text{MC} = PQ/V$.

The locking ratio. Define λ as the proportion of total supply that is locked or staked:

$$\lambda = \frac{M_{\text{locked}}}{M}, \qquad \lambda \in [0, 1).$$

The free-circulating supply is $M(1 - \lambda)$. We define the *free velocity* V_{free} as the velocity of only the economically active (unlocked) portion:

$$V_{\text{free}} = \frac{PQ}{M(1 - \lambda)}.$$

MoE and SoV components. The Medium of Exchange component of market capitalization is the value attributable to current transactional activity, priced at the free velocity:

$$\text{MC}_{\text{MoE}} = \frac{PQ}{V_{\text{free}}} = M(1 - \lambda).$$

The Store of Value component is the residual:

$$\text{MC}_{\text{SoV}} = \text{MC} - \text{MC}_{\text{MoE}} = M - M(1 - \lambda) = M\lambda.$$

The SoV/MoE ratio. The ratio of the SoV to MoE components is:

$$\frac{\text{MC}_{\text{SoV}}}{\text{MC}_{\text{MoE}}} = \frac{M\lambda}{M(1 - \lambda)} = \frac{\lambda}{1 - \lambda}.$$

This result shows that the SoV/MoE ratio is a monotone increasing function of λ alone, requiring no estimation of velocity, transaction volume, or any other derived quantity. As $\lambda \to 0$, the ratio approaches zero (pure medium of exchange). As $\lambda \to 1$, the ratio diverges (pure store of value).

The Growth-Levelized NVT ratio. The Network Value to Transactions ratio is NVT = $\text{MC}/PQ_{\text{annualized}}$. We construct a forward-looking version by replacing current PQ with PQ^*, the levelized present value of the projected transaction stream:

$$PQ^* = \frac{\sum_{s=1}^{n} \frac{PQ_0(1+g)^s}{(1+r_e)^s} + \frac{PQ_0(1+g)^n(1+g_\infty)}{(r_e-g_\infty)(1+r_e)^n}}{\text{annuity factor}(r_e, n)},$$

where g is the trailing k-year CAGR of PQ, r_e is the CAPM-estimated required return, g_∞ is the terminal growth rate, and the annuity factor converts the present value to a levelized annual equivalent (analogous to the Excel PMT function). The Growth-Levelized NVT ratio is then $\text{NVT}_{GL} = \text{MC}/PQ^*$. Full estimation details are in Section 4.

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